

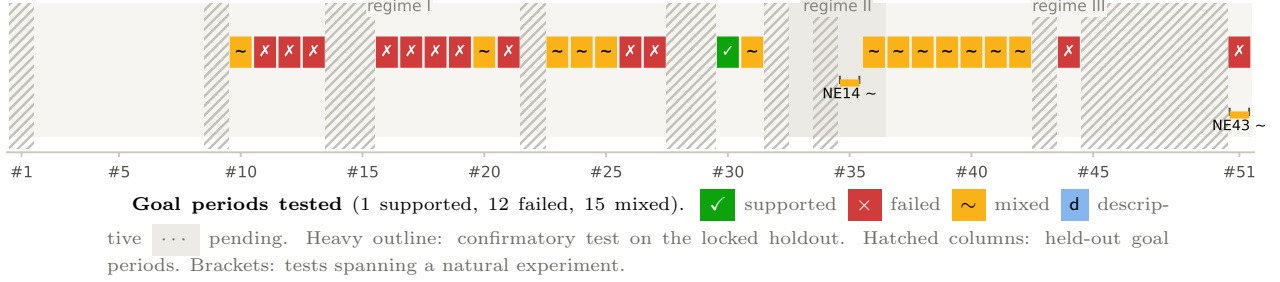
H49: Edge-trimmed regime III is a dilute ferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

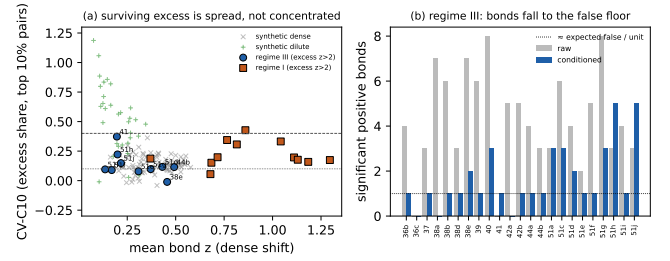
Bottom line. After trimming the operator's schedule, regime-III co-activation is not a dilute ferromagnet of small coupled clusters but a weak, structureless swarm-wide mode, with individual bonds at the false-positive floor.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Asked whether regime-III activity coupling, once H38's day-edge drive is trimmed, sits in a few strong pairs whose bond graph is below percolation. Inferred conditioned pairwise couplings in 23 regime-III and 23 regime-I period units, each against 200 block-shift surrogates calibrated on synthetic swarms at real schedules. **Refuted as posed.** Trimming removes 67% of regime-III bonds (17% in regime I). What is left is at the false-positive floor: 38 bonds against ≈ 23 expected false. The surviving collective excess (10 units, including #44 and #51) is a structureless shift of all pairs: the top 10% of pairs carry a median 11% of it, and pair structure doesn't replicate. #44's fine-tuning team appears in talk, not activity. Same-lab (provider-outage) pairs carry no excess bonds. At #51's sampling the method would have found strong bonds (recall 0.89). NE43: the edge drive outlives the daily bookend messages. Holdout script frozen, not run.



Observables. (a) Units with a significant edge-trimmed excess (regime III blue, regime I orange) against planted swarms at village sampling (grey $\times$ dense $J = 1.2/N$; green $+$ dilute $J = 0.5$). Real units sit with the dense cloud: low CV-C10, positive mean bond z . (b) Regime-III significant bonds per unit, raw (grey) vs trimmed (blue). Trimming takes them down to the ≈ 1 expected false bond per unit (dotted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a useful negative. If you run an always-on swarm and want to know who is coupled to whom from activity timing, the answer here is: nobody in particular. Once you trim the daily start and stop, there are no small cliques of agents that work in lockstep. What remains is a weak common mode shared by everyone, plausibly a shared fast drive rather than peer influence. So don't target pairs, and don't trust individual inferred bonds from 1-min activity. Monitor and steer the common mode, and look for real pairwise structure in who talks to whom, which is where the one known team showed up. The corrected NE43 date is a small data contribution.

PHYSICAL MODEL

Model 01 (inverse Ising). $s_i(t) = \pm 1$: agent i active in minute t . $\delta_i(b)$: agent field per 30-min block. x_j : block-centered spin with scaffold-unavailable minutes (not started, finished, consolidating, error gap) set to the block mean (H38's conditioning). J_{ij} : equal-time bond from pseudo-likelihood. A bond is significant when its surrogate z passes $p < 1/n_{\text{pairs}}$. Percolation: Molloy–Reed κ . CV-C10: cross-validated share of the excess covariance ΔC in the top 10% of pairs.

$$\text{logit } P(s_i(t) = 1 \mid \cdot) = 2 \left[\delta_i(b) + \sum_{j \neq i} J_{ij} x_j(t) \right], \quad x_j = a_j (s_j - \bar{s}_j^{(b)})$$

$$z_{ij} = (J_{ij} - \langle J_{ij} \rangle_{\text{shift}}) / \sigma_{ij}, \quad \text{bond if } p(z_{ij}) < 1/n_{\text{pairs}}$$

$$\kappa = \langle k^2 \rangle / \langle k \rangle \text{ (giant component iff } \kappa > 2), \quad \text{VR} - 1 = \sum_{i \neq j} \bar{C}_{ij} / \sum_i \bar{C}_i$$

H49-dilute-ferromagnet · generated 2026-10-04 11:10 by in-
fra/summaries/build_summaries.py · card hypotheses/H49-dilute-
ferromagnet/README.md · ratings: H49 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
Synthetic: calibrated null; dilute bonds recovered	size 0.054–0.062; recall 0.47 / 0.34 / 0.89	✓ / partial
Synthetic: percolation detectable	0.13 / 0.00 / 0.77 of percolating graphs	×
P1 trimmed bonds of pairs	$\leq 5\%$ 0.95%; 38 vs ≈ 23 false	✓
P2 excess concentrated (CV-C10 ≥ 0.4)	0/10 units; median (dense)	0.11 ×
P3 below percolation	23/23, trivially (power)	✓
P5 trimming removes bonds in III, not I	67% vs 17%	✓
NE43 edge drive bookends	$\times 0.77$, $\times 0.93$	✓
#44 team is a bond cluster	activity p 0.26, 0 bonds; talk p 0.013	×
#51 bonds recur; same-lab (R2)	r 0.02; OR 0.50	×
NE14 bonds persist II→III	no reliable pairs either side	n/a

GOAL PERIODS IN DETAIL

G10 $\sim$ mixed
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed
G17 $\times$ failed
G18 $\times$ failed
G19 $\times$ failed
G20 $\sim$ mixed
G21 $\times$ failed
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Equal-time inference is blind to coupling delayed by minutes, such as read-out-gated replies (H08). The verdict is about equal-time bonds. Dense J_0/N coupling and a fast shared drive look the same here; the post-hoc size law favours a field but relies on cross-period N. Individual bonds need $J \gtrsim 0.5$ and ≥ 5 days to be found. CV-C10 was redefined as a pooled ratio after the first pass, with no verdict change. All results were rebuilt after the activity-bins join bug (no verdict changed). The holdout has not been run.

ROUND 2

Lagged (read-out-lag) bonds; identify the shared field (human messages, nudges, provider latency) and test the size law at roster joins within #51; dilute-vs-dense in the talk channel, where the #44 team appears; run `confirm.py`.

- **Essence.** whether an agent swarm’s collective co-activation is carried by a few identifiable pairs (targetable) or by a swarm-wide mode (steer the field, not the pairs). For regime-III activity the answer is the mode.
- **H49-R1** Lagged bonds.
- **H49-R2** Identify the shared field.
- **H49-R3** Talk bonds.
- **H49-R4** Power-corrected percolation.
- **H49-R5** Run `confirm.py` (C1–C6).

Second figure. Planted swarms at three real schedules (30 replicates each). (a) Raw inference fakes bonds from the day edges; conditioning brings the null to the floor. (b) CV-C10 separates dense from dilute when the excess is significant. (c) Every structure reads as “below percolation”, even a percolating one. The concentration call is rare for dense coupling and common for dilute.

